Pascal's triangle

1

a $1, 5, 10, 10, 5, 1$

b 32

c ${}^5C_0, {}^5C_1, {}^5C_2, {}^5C_3, {}^5C_4, {}^5C_5$

2 $1, 8, 28, 56, 70, 56, 28, 8, 1$

3 7th

4

a 7

b 9

5

a 5

b 4

6 $(x+m)^4 = 1x^4m^0 + 4x^3m^1 + 6x^2m^2 + 4x^1m^3 + 1x^0m^4$

Binomial theorem

7

a 6

b 14

c 10

d 1

8

a $560b^3$

b $126c^5d^4$

c $135u^2v^4$

d $\frac{55}{128}u^3$

e $\frac{280}{3y^2}$

f $2288x^3y^{20}$

g $-3876x^4y^{75}$

h $1\,451\,520x^8y^{12}$

i $216x^{-6}y^{-4}$

j 1792

9 $45x^2y^8, 10xy^9, y^{10}$

10

a $n = 10$

b $45m^2k^8$

11 $p = 2$

13 ${}^{4}C_{3}$

14 $(4p+3q)^3 = {}^{3}C_0\,(4p)^3\,(3q)^0 + {}^{3}C_1\,(4p)^2\,(3q)^1 + {}^{3}C_2\,(4p)^1\,(3q)^2 + {}^{3}C_3\,(4p)^0\,(3q)^3$

15

a $256x^4$ **b** $81y^4$

16

a $\binom{5}{2}$ **b** 10 **c** 4th term

17 $55u^2v^9, 55u^9v^2$

18 $10\,777\,536$

19

a $1+4x+6x^2+4x^3+x^4$

b $32+80b+80b^2+40b^3+10b^4+b^5$

c $p^5+5p^4q+10p^3q^2+10p^2q^3+5pq^4+q^5$

d $c^8-8c^7d+28c^6d^2-56c^5d^3+70c^4d^4-56c^3d^5+28c^2d^6-8cd^7+d^8$

e $u^6+24u^5v+240u^4v^2+1280u^3v^3+3840u^2v^4+6144uv^5+4096v^6$

f $y^4+12y^3+54y^2+108y+81$

g $(ka)^6-6\,(ka)^5\,b+15\,(ka)^4\,b^2-20\,(ka)^3\,b^3+15\,(ka)^2\,b^4-6kab^5+b^6$

h $y^3-15y^2+75y-125$

i $y^3-y^2+\dfrac{y}{3}-\dfrac{1}{27}$

j $8+6y+\dfrac{3y^2}{2}+\dfrac{y^3}{8}$

k $27y^3+54ry^2+36r^2y+8r^3$

l $a^3-3a+\dfrac{3}{a}-\dfrac{1}{a^3}$

m $u^6+9u^4v^2+27u^2v^4+27v^6$

n $1+\dfrac{12}{y}+\dfrac{48}{y^2}+\dfrac{64}{y^3}$

20

a $405x^4$ **b** $120c^3d^7$ **c** $165u^3v^8$ **d** $3003u^5v^{10}$

21 $n=16$

$\frac{21}{2}$

23 $\binom{9}{3}$

24 9

25 ${}^{8}C_{k} \times 3^{8-k}\,(-x)^{k}$

26
a 0.943 **b** 1.037 **c** 0.890